

LESSON 9.7

DESCRIBING AND INTERPRETING HISTOGRAMS – PART 2



LESSON INTRODUCTION

MA.6.DP.1.4 Given a histogram or line plot within a real-world context, qualitatively describe and interpret the spread and distribution of the data, including any symmetry, skewness, gaps, clusters, outliers, and the range.

LEARNING TARGETS

- I can describe the shape of a histogram.
- I can describe the spread of a data set given a histogram.
- I can describe gaps, outliers, and clusters in a histogram.

BENCHMARK COHERENCE

BUILDING ON

MA.5.DP.1.2 Interpret numerical data with whole-number values, represented with tables or line plots, by determining the mean, mode, median, or range.

WORKING TOWARD

MA.7.DP.1.1 Determine an appropriate measure of center or measure of variation to summarize numerical data, represented numerically or graphically, taking into consideration the context and any outliers.

ESSENTIAL QUESTIONS

- How can you describe the distribution of a set of data on a histogram?
- What does a peak indicate? What about a gap?
- How do you determine a cluster? An outlier?

LESSON NARRATIVE

This lesson is Part 2 of 2 of interpreting histograms. Students begin by applying what they learned in Part 1 to describe the shape of a histogram and consider whether the shape is symmetrical or asymmetrical after reviewing the definitions. Students then examine several histograms and describe their symmetry, skew, and distribution. Students deepen understanding by analyzing the implications of any clusters, gaps, or outliers shown in a histogram.

COMPONENT SUMMARY

9.7.1 WARM-UP (~5 MINS)

Error analysis to determine a “math fail”

9.7.2 GUIDED INSTRUCTION (10 MINS)

Analyze the shape of a histogram to determine symmetry and skew

9.7.3 GUIDED PRACTICE (10 MINS)

Describe the distribution of data in a histogram

9.7.4 GUIDED INSTRUCTION (10 MINS)

Define and describe the features of a histogram

9.7.5 COLLABORATION (5 MINS)

Analyze the features of histograms to interpret and compare data sets

9.7.6 YOUR TURN! (5 MINS)

Analyze the shapes of two histograms to compare data

9.7.7 WRAP-UP (~5 MINS)

Self-assessment of describing and analyzing histograms and their features

RESOURCES

GLOSSARY

Key Terms:

- Right-skewed
- Left-skewed
- Cluster
- Gap
- Outlier

Additional Terms: N/A

MATERIALS

- (Optional) Colored pencils
- (Optional) Highlighters

ADDITIONAL PREPARATION

N/A

TEACHER TIPS

COMMON MISCONCEPTIONS

- Students may struggle to understand why the data on each side of a line of symmetry may not be exactly symmetrical.
- Students may confuse “left-skewed” and “right-skewed.”
- Students may struggle to determine if an interval refers to a distance on the x- or the y-axis.

THINGS TO CONSIDER

- Have students consider if the data on each side of a line of symmetry is generally symmetrical as opposed to exact.
- Encourage students to describe the distribution of data based on initial visual impressions instead of exact mathematical calculations.
- Have students notice the context in which terms, such as “interval,” are used.
- If time is an issue, then consider assigning Your Turn! for homework.

LITERACY AND LANGUAGE ROUTINES

Take Turns

9.7.5 Collaboration is an opportunity for students to engage in a Take Turns routine. This encourages students to work independently and then engage in mathematical discourse with a partner to explain and justify their responses. This is particularly effective in deepening student understanding and strengthening analytical skills.

MATHEMATICAL THINKING AND REASONING (MTR) STANDARD**MA.K12.MTR.5.1 Patterns and Structures**

Students will be working to describe the shape of a histogram and its data set. Clarifications within this standard emphasize recognizing patterns and applying these patterns to mathematical concepts. Utilize the progression of the lesson to highlight the process by which students use details within a problem to form generalizations about a data set.

DIFFERENTIATE INSTRUCTION

The display of data through the graphical representations on histograms provides visual entry points for students to engage with the content and begin to analyze the information presented. Additional visual entry points can be accessed through the use of colored pencils to label the features of the histograms to better visualize the story of the data.

9.7.1 WARM-UP: MATH FAIL

Component Overview: Students use an open-ended prompt to examine a retail sign. This activity encourages students to be discerning observers of real-life math.

TEACHER MOVES

Consider encouraging students to think about how this sign came to be. Allow students to demonstrate creativity with their answers. Use sentence stems to encourage conversation such as “I think...” or “I wonder...”

**9.7.1 Warm-Up: Math Fail**

1. Discuss with your partner what you notice about this picture.

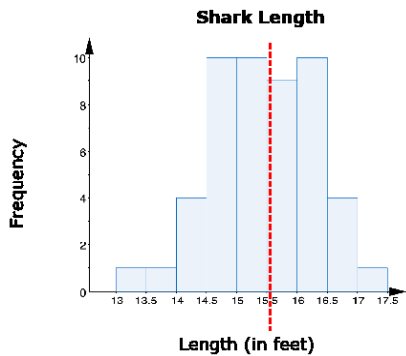


The price being advertised is actually more than the original price.



9.7.2 Guided Instruction: Shape of Distributions

1. Consider the histogram of the lengths of female sharks.



- Sketch an approximate line of symmetry for the histogram.
Answers will vary.
- If the histogram is folded along the line of symmetry, will each half line up perfectly?
No, the histogram will not line up perfectly.
- Do you think all histograms will be symmetrical?
Answers will vary. No, I do not think all histograms can be described as symmetrical.

9.7.2 GUIDED INSTRUCTION: SHAPE OF DISTRIBUTIONS

Component Overview: Students describe the shape of a distribution displayed in a histogram. Students consider if a histogram is symmetric or asymmetric as well as if it skews to the right or to the left. Encourage students to consider how the shape of a distribution allows them to make generalizations or draw conclusions about a data set.

TEACHER MOVES

Have students compare and contrast the exact symmetry of the heart with the general symmetry of the histogram. Encourage students to describe if they think that the term “symmetrical” can still be used for things that are not exactly symmetrical. Challenge students to think of real-life items that can be generally, but not exactly, symmetrical.

DIFFERENTIATE INSTRUCTION

This section provides visual entry points through the graphical display of data on the histogram. Auditory entry points are provided through partner discussion. Consider additional kinesthetic entry points to engage all learners.

9.7.2 GUIDED INSTRUCTION: SHAPE OF DISTRIBUTIONS (CONTINUED)

ENRICHMENT

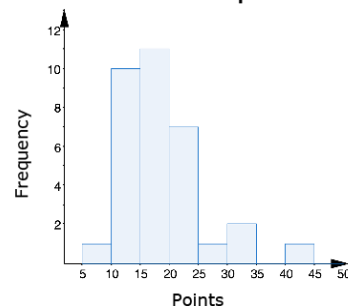
Synthesize student learning through the use of intentional questioning.

- Many words have the prefix “a-”: asocial, anonymous, apolitical, asymmetric. What does this prefix mean?
- How does it help you understand the meaning of “asymmetric”?
- How would you describe the appearance of the data in the histograms without using the words “symmetric” or “asymmetric”?

The shape of the shark lengths histogram can be described as **symmetric**, but histograms can also have other shapes.

2. The histogram displays the number of points per game scored by a college basketball player during the 2008 season.

2008 Points per Game



Discuss with your partner what you notice about the shape of the distribution of points scored per game.

Students do not need to write an explanation here, but a sample response is provided for reference. Most of the data in this set is on the left side of the histogram. There are fewer large values in this data set compared to small values.

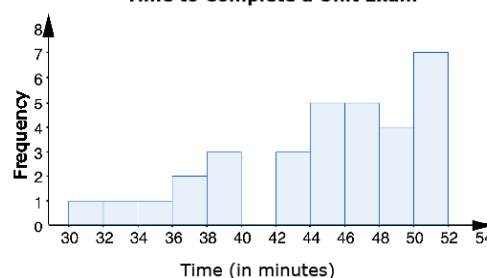
A **right-skewed** graph has a longer trail of data to the right and the peak of the data on the left.

TEACHER MOVES

Ask students what they think the word “skew” means outside of a mathematical context. Encourage students to think of examples when they might have heard the word “skew” and brainstorm contexts in which they could use the word in their everyday lives. Connect this discussion to the description of histograms as either being left-skewed or right-skewed.

3. The time to complete a unit exam for 32 students is shown on the histogram.

Time to Complete a Unit Exam



Write an observation about the shape of the distribution of time spent to complete the exam.

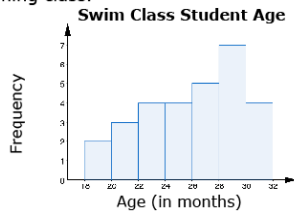
Answers will vary. Most of the data in this data set is on the right side of the histogram with less data trailing on the left side.

A **left-skewed** graph has a longer trail of data to the left and the peak of the data on the right.



9.7.3 Guided Practice: Describing Shape of Distributions

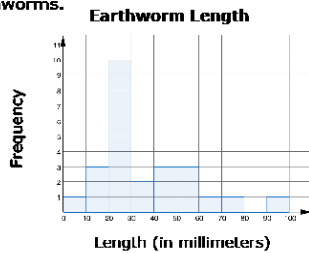
1. This histogram shows the ages of students in a swimming class.



The shape of the distribution of ages can be best described as

- ☐ symmetric.
- ☐ right skewed.
- ☒ left skewed.

2. The histogram shows the length, in millimeters, of 25 earthworms.



The shape of the distribution of the earthworms' lengths can be best described as

- ☐ symmetric.
- ☒ right skewed.
- ☐ left skewed.

9.7.3 GUIDED PRACTICE: DESCRIBING SHAPE OF DISTRIBUTIONS

Component Overview: Students describe the shape of the distribution of histograms. Students use the distribution of a histogram to make inferences and draw conclusions about a data set. Encourage students to consider how the data reflects real-life situations.

TEACHER MOVES

This section is intentionally scaffolded through the use of multiple choice to describe the skew of histograms. Encourage students to refer to the previous section if they are having trouble determining the direction of the skew.

To provide additional scaffolding, encourage students to draw a vertical line down the center of the histogram and describe what they see on each side. This will provide a visual for students to connect to their descriptions.

ENRICHMENT

Provide additional scaffolding by asking guiding questions.

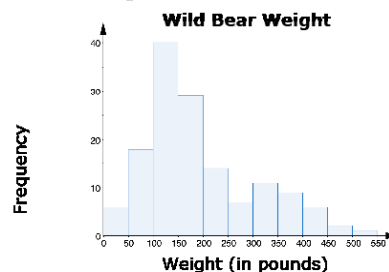
- Where is the center of the histogram? Are there more data points to the left or right of it?
- Where is the peak of the histogram? What does this tell you about skew?

9.7.3 GUIDED PRACTICE: DESCRIBING SHAPE OF DISTRIBUTIONS (CONTINUED)

DIFFERENTIATE INSTRUCTION

Consider having students use colored pencils or highlighters to mark or note areas of interest on the histograms. Some students may also benefit from drawing a horizontal line from the top of each bar to its corresponding value on the vertical axis.

3. In one study of wild bears, researchers measured the weights, in pounds, of 143 wild bears that ranged in age from newborn to 15 years old. The data were used to make this histogram.



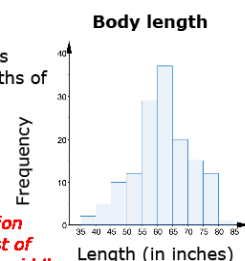
Describe the shape of the graph given the context.

The shape of the Wild Bear Weight histogram is right-skewed. Most of the bears weighed between 50 and 250 pounds, but the range goes all the way up to almost 550 pounds.

4. The histogram summarizes the data on the body lengths of the same 143 wild bears.

Describe the shape of the distribution of the lengths of bears.

The shape of the distribution is roughly symmetric. Most of the data values are in the middle of the distribution.

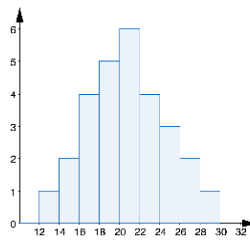




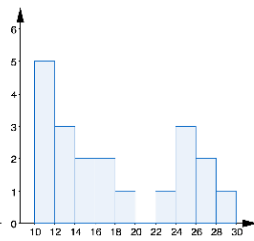
9.7.4 Guided Instruction: Features of Histograms

Below are two histograms with different shapes and features.

Histogram A



Histogram B



1. Histogram A is ☐ symmetrical and has a peak near 20.
☐ not symmetrical
2. Histogram B is ☐ symmetrical and has ☐ not symmetrical.
☐ one peak.
☐ more than one peak.



A **cluster** is formed when data points are in a close group or numerical data have similar values on a number line.

3. ☐ Histogram A has two clusters.
☐ Histogram B

9.7.4 GUIDED INSTRUCTION: FEATURES OF HISTOGRAMS

Component Overview: Students consider the features of histograms. Students use the terms “peak,” “cluster,” “gap,” and “outlier” to describe a histogram and make inferences about the data set represented.

TEACHER MOVES

Invite students to consider when they have heard or used each word—cluster, gap, peak, outlier—in their daily lives. With a partner, have students discuss the everyday meanings and the mathematical meanings of the terms. Encourage students to note any overlap in the meanings.

DIFFERENTIATE INSTRUCTION

Have students fold a piece of paper into four equal sections and then label each section with one of the following terms: cluster, gap, peak, and outlier. In the corresponding section, have students write the mathematical definition of the term and draw an illustration depicting it.

9.7.4 GUIDED INSTRUCTION: FEATURES OF HISTOGRAMS (CONTINUED)

ENRICHMENT

To encourage conceptual understanding over memorization, ask questions like:

- Why do you think outliers occur? What would be an outlier in a histogram showing the heights of students in our class?
- How could you determine that a number is an outlier? Are outliers dependent on the situations they depict? Why or why not?

A **gap** shows a location with no data values. On a histogram, a space between the bins indicates a gap, or a frequency of zero for that bin.

4.
☐ Histogram A
☒ Histogram B
 has a gap between 20 and 22.



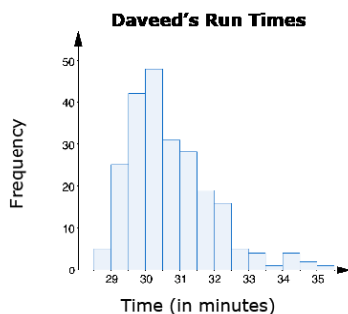
An **outlier** is a value that is much higher or much lower than the other values in a data set.

5. If the data value
☐ 12
☐ 24
☒ 40
 were added to Histogram A, it would be considered an outlier.



9.7.5 Collaboration: Features of Histograms

1. Daveed has been tracking the time, in minutes, it takes him to run 4 miles for the last 10 years. A histogram of Daveed's run times is shown.



- a. The shape of run times can be best described as
- ☐ symmetrical.
☒ right skewed.
☐ left skewed.
- b. Daveed's typical run time was about 30 minutes. Most of his run times were between 29.5 to 31 minutes. Not many of his runs were ☐ shorter ☒ longer than 32.5 minutes.
- Answers for the first blank may vary.*

9.7.5 COLLABORATION: FEATURES OF HISTOGRAMS

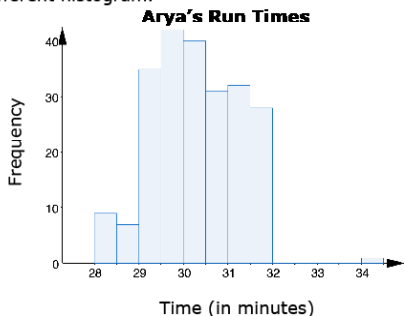
Component Overview: Students can summarize or synthesize their findings in this activity. Partners will work together on all of the items or each solve half of the problems and then share their answers. Also consider placing students in small groups, with each student solving a problem and then using a jigsaw method to share their findings.

TEACHER MOVES

Take Turns

Consider using a Take Turns routine for the Collaboration. Students will take turns answering each question. The student who does not answer a particular question will explain their partner's answer in their own words. If a partner answers or explains inaccurately, the pair can then work together to solve the problem.

2. Arya used the data of her 4-mile run times to create a different histogram.



- a. Discuss with your partner what you notice about the interval widths. Determine if there are any unusual features including clusters, gaps, or outliers. Write a detailed description of your discussion.
- Answers will vary. The x-axis begins at 28 minutes and then increases by 0.5. There is a gap in Arya's run times from 32 to 34. The run time between 34 and 34.5 minutes could be an outlier since it is not close to the other data values. Most of Arya's runs last between 29 to 31.5 minutes.*
- b. At what interval in the histogram of Arya's times is there an outlier?
- Between 34 to 34.5 minutes.*
- c. Would this same data value also be considered an outlier in the histogram of Daveed's times?
- No, Daveed's has runs that are between 34 to 35 minutes, and runs that are even longer.*

TEACHER MOVES

To guide student discussion and provide scaffolding support for students who may struggle with the partner discussion, ask guiding questions like:

- What do you notice about the intervals? What does this tell you about the data set?
- Are there any gaps in the data? What does this mean for outliers?
- How can you describe Arya's run times? What data backs up this statement?

9.7.5 COLLABORATION: FEATURES OF HISTOGRAMS (CONTINUED)

TEACHER MOVES

MA.K12.MTR 5.1: Patterns and Structures

Invite students to recognize patterns in the histograms in this activity and then draw inferences or form generalizations about the individuals running based on the histograms.

DIFFERENTIATE INSTRUCTION

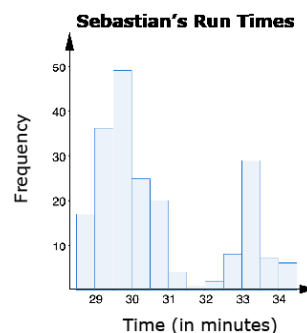
Allow students to verbally explain their answers to the questions. Some students struggle to convey their words in writing but can offer detailed verbal observations. Both written and verbal answers are valid.

ENRICHMENT

Students may struggle to compare how the data sets impact the features of the histograms. If you notice partner pairs struggling to engage in discourse, ask questions to provide scaffolding:

- How are the histograms representing Daveed's, Arya's, and Sebastian's run times similarly? How do they differ?
- What summary statement could you make about Daveed's run times? Arya's? Sebastian's?

3. Sebastian used the data of his 4-mile run times to create a third histogram.



- a. Which feature is depicted in Sebastian's histogram?

Answers will vary. Sebastian has two clusters of data values, with two peaks. One peak is between 29.5 and 30 and the other is between 33 and 33.5.

- b. Describe the spread and shape of the distribution of Sebastian's run times.

Answers will vary. Sebastian's run times has two peaks and the data took a dip between 31 and 32.5 minutes. Perhaps the first cluster of runs are at a time of day that he has more energy and the second set is at a time when he is more tired.

4. Different data will change the intervals for the horizontal axis and will affect the histogram and its features: symmetry, skewness, gaps, clusters, outliers, and the range. With your partner, review all three run-time scenarios and discuss the differences between the histograms.

Answers will vary. Students do not need to write an explanation here, but a sample response is provided for reference.

Daveed: This data set is right skewed. Daveed's run times range from 28.5 to 35.5 minutes.

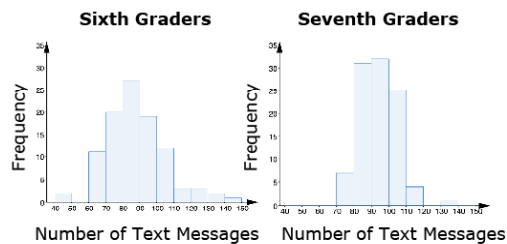
Arya: This data set has a gap on the right side on the graph from 32 to 34 minutes. Her run that took more than 34 minutes looks like an outlier since no other runs took that long.

Sebastian: This data set has two peaks. He has a low frequency between 31.5 to 32 minutes.



9.7.6 Your Turn!

1. The histograms show the number of text messages sent in one week by a group of 100 sixth-grade and 100 seventh-grade students.



- Describe the shape of the histogram for the number of text messages sent in one week by sixth-grade students.
Answers will vary. The sixth-grade data is roughly symmetric.
- Which group of students has the larger spread?
Sixth-Grade Students.
- Explain if sixth graders or seventh graders tend to be more alike in terms of the number of text messages they send.
Answers will vary. Seventh graders are more consistent because the data has a smaller range and the data values are closer together.

9.7.6 YOUR TURN!

Component Overview: In this Your Turn!, students compare text message data from 6th and 7th graders using histograms. Students analyze the histograms to describe their shape and determine which class has greater spread and which class is more alike in their cell phone usage.

ENRICHMENT

If you notice students struggling to complete this section independently, ask questions to further scaffold:

- Would you describe these histograms as symmetrical or asymmetrical? Why? What does that tell you about the data?
- How can you determine which grade has the largest spread?
- Will the class with the largest spread be more or less alike in their cell phone usage? Why?

9.7.7 WRAP-UP: REFLECTION

Component Overview: Students reflect on how they feel about describing the shape of a histogram using mathematical language. Consider using this data to inform misconceptions to address in a following lesson or prioritize students to check in with before moving forward with new content.

TEACHER MOVES

Encourage students to explain their responses in as much detail as possible using specific examples from the lesson. If needed, allow students to use sentence stems such as:

- “I feel comfortable with...”
- “One question I still have is...”
- “An activity that I struggled with was...”



9.7.7 Wrap-Up: Reflection

Place an **X** on the line to indicate your understanding of parts of analyzing a histogram. Then provide a brief explanation or evidence to support your self-assessment.

Answers will vary.

1. I can describe the shape of a histogram.

I need help.
I can't get started.

I'm getting there.

I understand
and have evidence.



2. I can describe gaps, outliers, and clusters in a histogram.

I need help.
I can't get started.

I'm getting there.

I understand
and have evidence.



3. I can describe the spread of a data set given a histogram.

I need help.
I can't get started.

I'm getting there.

I understand
and have evidence.



